

Abstract No.

Dexamethasone induces immunosuppressive remodeling of the brain tumor microenvironment associated with poor prognosis

Laurens Zschke^{1,2,}, Niklas Rutsch¹, Maleen Rehberger¹, Lukas Klein¹, David Wasilewski³, Bjarne Dänekas¹, Johannes Reilly¹, Cihat Karadag¹, Peter Vajkoczy¹, Daniel Teichmann¹, MOSAIC Consortium⁴, David Capper¹, Ilon Liu¹, and Julia Onken¹*

¹Charité - University Hospital Berlin

²Berlin Institute for Medical Systems Biology

³Universitätsklinikum Düsseldorf

⁴Owkin Inc.

*Presenting author

Brain tumors patients often develop brain edema which is routinely treated with dexamethasone (DEXA). We previously demonstrated that preoperative DEXA administration is an independent negative prognostic factor in both glioblastoma (GB) and non-small cell lung cancer brain metastases (NSCLCm). Although glucocorticoids are known to suppress inflammatory responses, the cellular and molecular mechanisms underlying their long-lasting effects on the brain tumor microenvironment remain poorly understood.

To address this question, we analyzed single-nucleus RNA sequencing data from 89 GB, including 74 patients from the MOSAIC consortium, and 30 NSCLCm specimens obtained at primary surgery and stratified samples according to preoperative DEXA exposure (none, short-term, or long-term). To extend the analysis longitudinally, we additionally sequenced 15 matched recurrent tumor samples.

Preliminary analyses identified multiple DEXA-associated immunosuppressive programs in myeloid cells, including upregulation of IL1R2 in neutrophils and CD163 in microglia, accompanied by reduced T-cell activation characterized by decreased IFN- γ signaling and diminished CD40–CD40L interactions. The magnitude of these immunosuppressive signatures correlated with expression of canonical glucocorticoid receptor (GR) target genes, including FKBP5 and ZBTB16. Notably, both baseline GR target gene activity and the transcriptional response to DEXA varied substantially between patients. Ongoing studies are investigating whether GR-pathway activation measured in peripheral blood correlates with GR-dependent remodeling of the tumor microenvironment.

Together, these findings provide a potential mechanistic explanation for the adverse clinical effects associated with DEXA treatment in brain tumor patients and suggest that baseline GR activity may serve as a biomarker to guide personalized glucocorticoid therapy.



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